

30/04/25

Lecture 29

The S^4 model (ϕ^4 theory??)

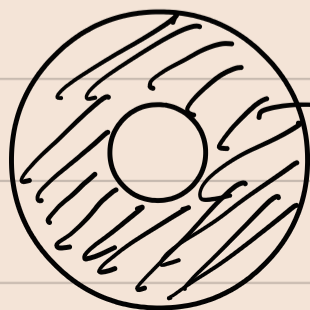
$$K(\sigma) = \underbrace{K_F(\sigma)}_{\text{free / quadratic hamiltonian}} + \underbrace{K_I(\sigma)}_{\text{Interaction Part}}$$

$$K_F(\sigma) = -\frac{1}{2} \int_q (q^2 + \tau) \sigma_q \sigma_{-q} d\sigma_q d\sigma_{-q}$$

$$K_I(\sigma) = -u \iiint_{q_1, q_2, q_3} \sigma_{q_1} \sigma_{q_2} \sigma_{q_3} \sigma_{-q_1} \sigma_{-q_2} \sigma_{-q_3}$$

$$Z = \int_{\sigma_0} e^{K_F(\sigma_0)} \int_{\sigma_1} e^{K_F(\sigma_1) + K_I(\sigma_0, \sigma_1)}$$

We will expand K_I & evaluate the σ_1 trace. Keeping only upto 2 orders in K_I .



Integrate out $|q| > \frac{1}{2}$

We will later justify this for $d=4$ or higher & then take $d=4-\varepsilon$ expansion.

$$Z = \int_{\sigma_0} e^{K_F(\sigma_0)} \int_{\sigma_1} e^{K_F(\sigma_1)} \left[1 + K_I(\sigma_0, \sigma_1) + \frac{K_I^2(\sigma_0, \sigma_1)}{2!} \right]$$

$$= \left[\int_{\sigma_0} e^{K_F(\sigma_0)} \int_{\sigma_1} e^{K_F(\sigma_1)} \right] \left[1 + \frac{\int_{\sigma_1} e^{K_F(\sigma_1)} K_I}{\int_{\sigma_1} e^{K_F(\sigma_1)}} + \frac{\int_{\sigma_1} e^{K_F(\sigma_1)} K_I^2}{2! \int_{\sigma_1} e^{K_F(\sigma_1)}} + \dots \right]$$

$$= \left[\int_{\sigma_0} e^{K_F(\sigma_0)} \int_{\sigma_1} e^{K_F(\sigma_1)} \right] \left[1 + \langle K_I \rangle_1 + \frac{\langle K_I^2 \rangle_1}{2!} + \frac{\langle K_I^3 \rangle_1}{3!} + \dots \right]$$

$$\langle K_I \rangle_1 = \frac{\int_{\sigma_1} e^{K_F(\sigma_1)} K_I}{\int_{\sigma_1} e^{K_F(\sigma_1)}}$$

$$Z = \int_{\sigma_0} \int e^{K_F(\sigma_0)} \left[\int_{\sigma_1} \int e^{K_F(\sigma_1)} \underbrace{\left\langle e^{K_I(\sigma_0, \sigma_1)} \right\rangle_1}_{\text{depends on } \sigma_0} \right]$$

upto 2nd order we can write

$$Z = \int_{\sigma_0} \int e^{K_F(\sigma_0)} \int_{\sigma_1} \int e^{K_F(\sigma_1)} \left[\left\langle K_I \right\rangle_1 + \frac{1}{2} \left[\left\langle K_I^2 \right\rangle_1 - \left\langle K_I \right\rangle_1^2 \right] \right]$$

$$Z = \int_{\sigma_0} \int e^{K_F(\sigma_0)} e^{\left\langle K_I \right\rangle_1 + \frac{1}{2} \left[\left\langle K_I^2 \right\rangle_1 - \left\langle K_I \right\rangle_1^2 \right]} \times \int_{\sigma_1} \int e^{K_F(\sigma_1)}$$

$$Z \propto \int_{\sigma_1} \int e^{\mathcal{K}'(\sigma_1)}$$

$$\mathcal{K}'(\sigma_1) = K_F(\zeta \sigma_{2q}^1) + \left\langle K_I(\zeta \sigma_{2q}^1, \sigma_1) \right\rangle_1 + \frac{1}{2} \left\{ \left\langle K_I^2(\zeta \sigma_{2q}^1, \sigma_1) \right\rangle_1 - \left\langle K_I(\zeta \sigma_{2q}^1, \sigma_1) \right\rangle_1^2 \right\}$$

From,

$$\mathcal{K}(\sigma_a^1) = \mathcal{K}_F(\mathcal{G}\sigma_{2q}^1) + \langle \mathcal{K}_I(\mathcal{G}\sigma_{2q}^1, \sigma_i) \rangle_1 \\ + \frac{1}{2} \left\{ \langle \mathcal{K}_I^2(\mathcal{G}\sigma_{2q}^1, \sigma_i) \rangle_1 - \langle \mathcal{K}_I(\mathcal{G}\sigma_{2q}^1, \sigma_i) \rangle_1^2 \right\}$$

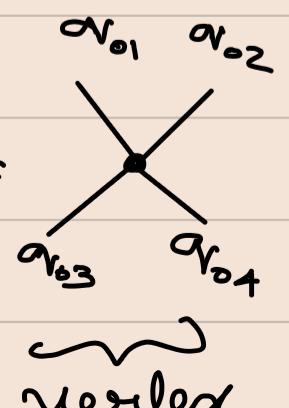
Our task is to evaluate

$$\langle \mathcal{K}_I(\mathcal{G}\sigma_{2q}^1, \sigma_i) \rangle_1 \text{ \& } \langle \mathcal{K}_I^2(\mathcal{G}\sigma_{2q}^1, \sigma_i) \rangle_1$$

$$\mathcal{K}_I(\sigma) = -\mathcal{U} \iiint_{q_1, q_2, q_3, q_4} \sigma_{q_1} \sigma_{q_2} \sigma_{q_3} \sigma_{q_4} \delta(q_1 + q_2 + q_3 + q_4)$$

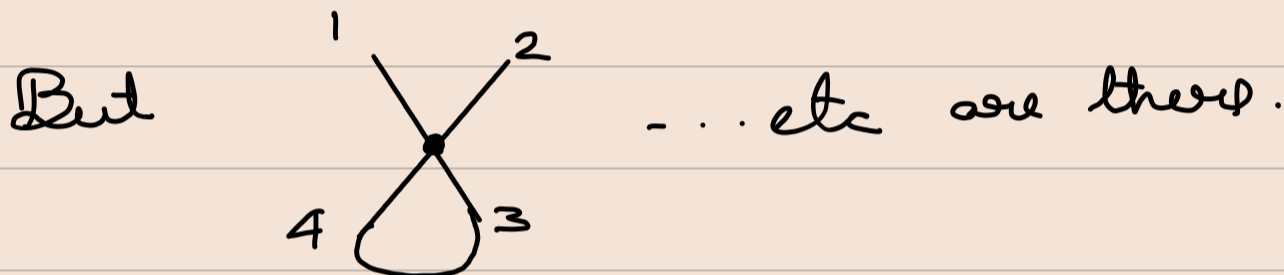
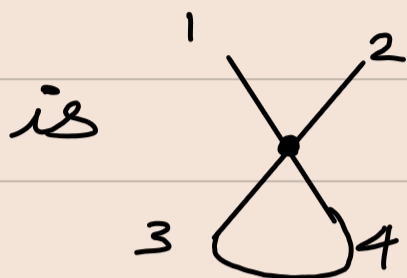
Now these σ_{q_i} have both $\sigma_{q_{0i}}$ & $\sigma_{q_{1j}}$ in them we should do integration over σ_{q_1} 's only.

A term like

$$\mathcal{U} \iiint_{1, 2, 3, 4} \sigma_{q_{01}} \sigma_{q_{02}} \sigma_{q_{03}} \sigma_{q_{04}} \delta(q_{01} + q_{02} + q_{03} + q_{04}) =$$


A term like

$$2 \int_1^2 \sigma_{q_{01}} \sigma_{q_{02}} \delta(q_{01} + q_{02}) \int_3^4 \sigma_{q_{13}} \sigma_{q_{14}} \delta(q_{13} + q_{14})$$



So there are 6 terms in there

And a term like

$$-2 \int_1^2 \int_3^4 \int_1^2 \int_3^4 \sigma_{1q_1} \sigma_{1q_2} \sigma_{1q_3} \sigma_{1q_4} \delta(\dots) \Rightarrow$$

And there are 3 terms like this

These terms appear in

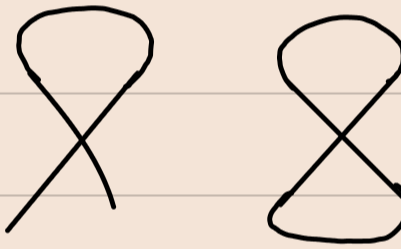
$\langle K_I \rangle_1$. & all the above are 1st order

2nd order

(i) Disconnected diagrams appear in $\langle K_I^2 \rangle_1$ & also in $\langle K_I \rangle_1^2$.
 & they cancel since

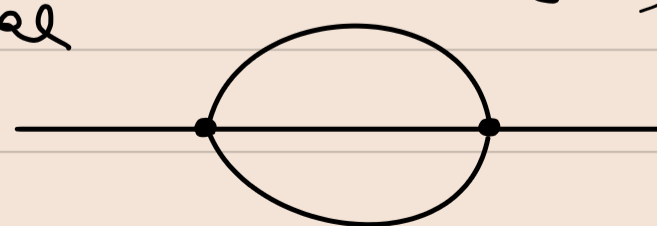
$$\langle K_I^2(\mathcal{G}_{2,1}^1, \sigma_1) \rangle_1 = \langle K_I(\mathcal{G}_{2,1}^1, \sigma_1) \rangle_1^2$$

appear in $K_I'(\sigma_1)$.

Eg:-  $\Rightarrow$ multiplying two integrals with no common variable b/w them

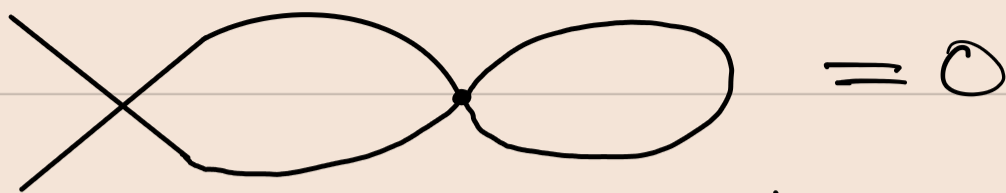
So $\left\{ X + 6 \text{ (figure-eight)} + 3 \text{ (figure-eight with line)} \right\}^2$ also appears in $\langle K_I^2 \rangle$.

$\underbrace{\hspace{10em}}_{\langle K_I \rangle^2}$

(ii) The diagram like  (Saturn diagram)

appears 48 times but its value is zero as it has odd pairings.

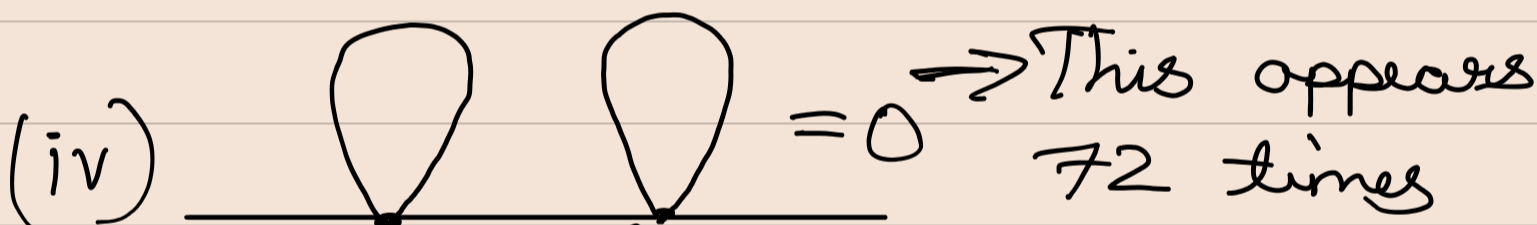
(iii) The diagram like



appears 72 times

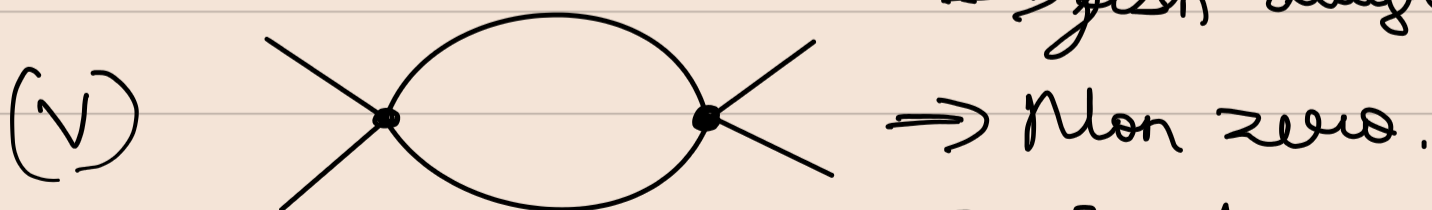
It is a 2nd order contribution to δ (2-free ends, so).

Thus we neglect this.



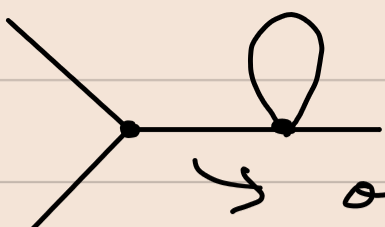
but zero because of one odd pairing

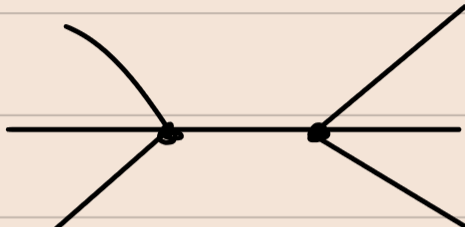
→ fish diagram



& contributes to δ term

& appears 36 times

(v) $48 \times$  $= 0$
 $\rightarrow$ odd pairing

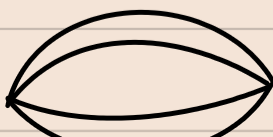
(vi) $8 \times$ 

Contributes to S^6 term
 but it doesn't matter for the
 fixed points that we are
 interested in (due to S^4 interaction)

How we get the value
 of integrals

(vii) $36 \times$ 

Constant ∇ doesn't
 contribute to recursion
 relation $\swarrow$

(viii) $12 \times$ 

The recursion relation is

$$\tau' = 4 \left[\tau + 3c \left(\frac{u}{1+\tau} \right) \right]$$

gaussian part $\swarrow$ got by the integral values.

$$\& \quad u' = 2^{4-d} \left\{ u - 9c \left[\frac{u^2}{(1+\tau)^2} \right] \right\}$$

$\downarrow$
appears from
Spin relabelling of σ .

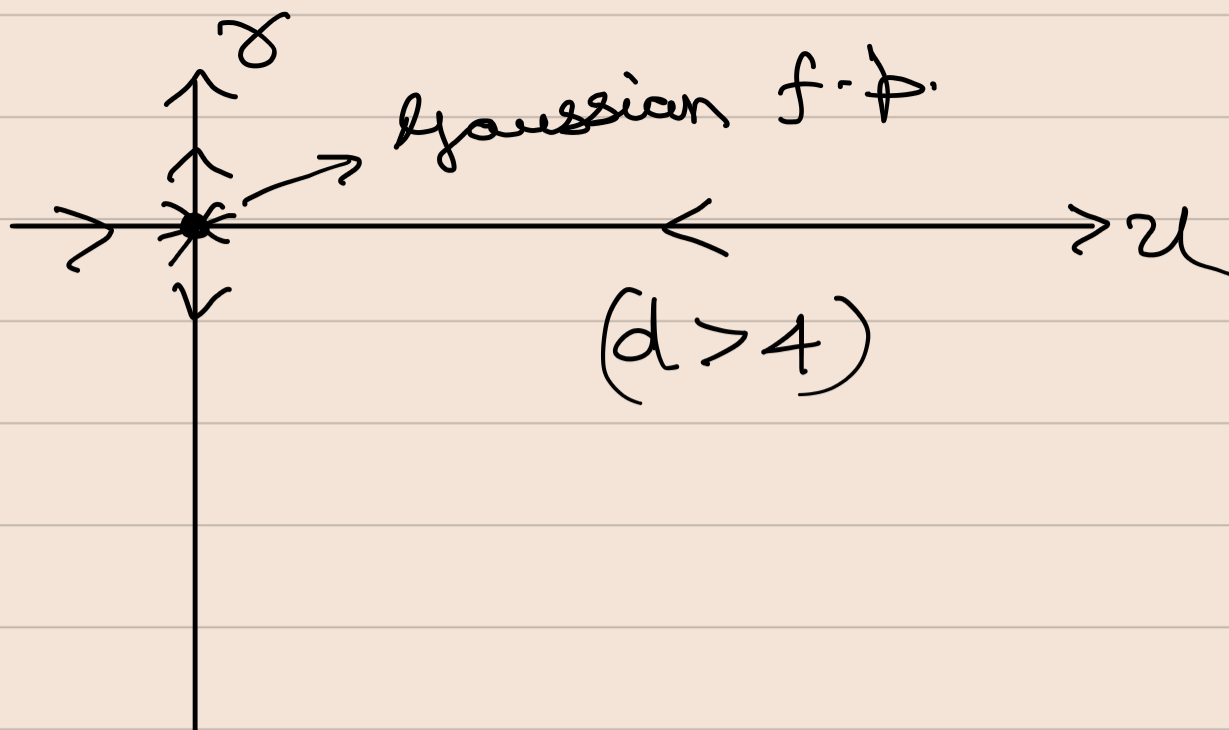
Thus if $d > 4$.

$\Rightarrow$ u becomes irrelevant

& the model becomes Gaussian.

Thus Mean field theory is exact
for $d > 4$.

$u \rightarrow$ irrelevant at Gaussian f.p.



Thus Gaussian f.p. is stable w.r.t u -axis.

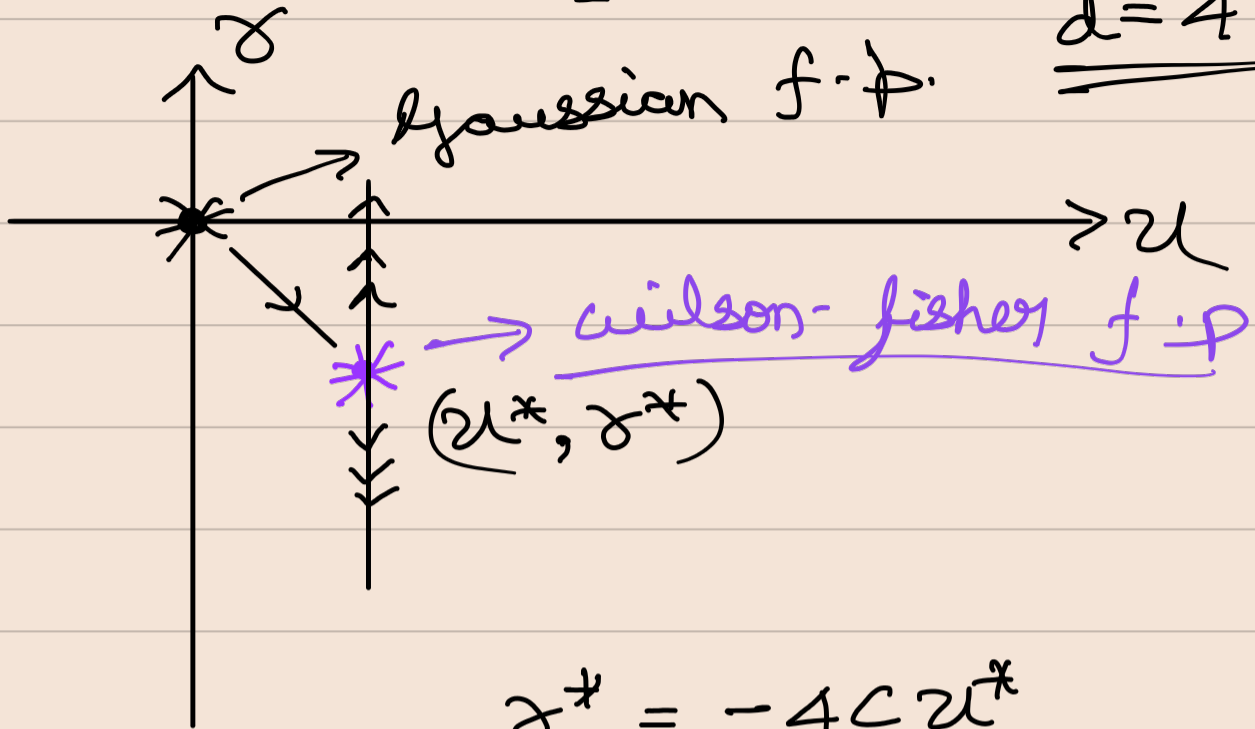
& in u -axis $\sigma' = 4\sigma$.

$$\text{Mean field} \left\{ \begin{array}{l} \frac{\partial \sigma'}{\partial \sigma} \Big|_{\sigma^*} = 4 = 2^{\lambda_L} \\ \lambda_L = 2. \\ \Rightarrow \nu = 1/2 \end{array} \right.$$

For $d > 4$

Now ' u ' is relevant at Gaussian f.p. & RG flow goes away from Gaussian f.p. to another f.p.

$d < 4$ let $d = 4 - \varepsilon$



$$z^* = -4C u^*$$

$$\& u^* = 1 + \varepsilon \ln(2) \left[u^* - 9C u^{*2} \right]$$

$\rightarrow u^*$ in leading order of ε

is
$$u^* = \frac{\varepsilon \ln(2)}{9C}$$

$$\& z^* = -\frac{4}{9} \varepsilon \ln(2)$$

Linearizing around $\sigma^* \neq u^*$

$$\begin{bmatrix} \delta \sigma \\ \delta u \end{bmatrix} = \begin{bmatrix} 4 - \frac{4\varepsilon \ln(2)}{3} & 12c(1-\sigma^*) \\ 0 & [1 - \varepsilon \ln(2)] \end{bmatrix} \begin{bmatrix} \delta \sigma \\ \delta u \end{bmatrix}$$

$$\Lambda_1 = 4 - \frac{4}{3} \varepsilon \ln(2)$$

$$\Lambda_1 = 2^{\lambda_1}$$

$$\lambda_1 = 2 - \frac{1}{3} \varepsilon$$

$$\Lambda_2 = 1 - \varepsilon \ln(2)$$

$$\lambda_2 = -\varepsilon$$

} all in
leading order
of ' ε '

$$\gamma = \frac{1}{\lambda_1} = \frac{1}{2} + \frac{\varepsilon}{12}$$

$$\text{for } d=3, \varepsilon=1 \rightarrow \frac{1}{2} + \frac{1}{12} = \underline{\underline{0.583}}$$

but exact ans. for $d=3 \sim \underline{\underline{0.623}}$